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Automatic snapshot selection and zone creation

Abstract

Aspects of the present disclosure involve a system comprising a computer-readable storage medium storing a program and method for automatic snapshot selection and zone creation. The program and method provide for receiving, from a first device, an indication of user input selecting to determine zoning metrics for a webpage based on a selected device type; accessing plural session replays corresponding to different visitor sessions by at least one second device with respect to the webpage; selecting a session replay based at least in part on maximizing scroll amount associated with the session replay; selecting, from among plural available snapshots corresponding to the selected session replay, a snapshot based at least in part on minimizing display of additional elements on the webpage; automatically identifying, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage; and determining zoning metrics for display on the first device.

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Background/Summary

CLAIM OF PRIORITY (1) This Application claims the benefit of priority of U.S. Provisional Application No. 63/580,310, filed Sep. 1, 2023, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD
(1) The present disclosure relates generally to web session analysis, including automatic snapshot selection and zone creation.
BACKGROUND
(2) Web analysis solutions provide for the collection and analysis of website data. Such solutions may provide for capturing user interaction with respect to webpage visits.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some non-limiting examples are illustrated in the figures of the accompanying drawings in which:

(2) FIG. **1** is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

(3) FIG. **2** is a diagrammatic representation of an experience analytics system, in accordance with some examples, that has both client-side and server-side functionality.

(4) FIG. **3** is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

(5) FIG. **4** illustrates an unlabeled document object model (DOM) tree, in accordance with some examples.

(6) FIG. **5** is a user interface for presenting a webpage with performance information for zones in a webpage, in accordance with some examples.

(7) FIG. **6** illustrates an architecture for automatic snapshot selection and zone creation, in accordance with some examples.

(8) FIG. **7** is a flowchart illustrating a process for automatic snapshot selection and zone creation, in accordance with some examples.

(9) FIG. **8** is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, in accordance with some examples.

(10) FIG. **9** is a block diagram showing a software architecture within which examples may be implemented.

DETAILED DESCRIPTION

(11) Web analysis solutions provide for the collection and analysis of website data. Example web analysis tools include the tracking and recording of session events corresponding to user interactions, automated website zone identification, session replay, statistical analysis of collected data, and the like.

(12) The disclosed embodiments as described herein provide for an experience analytics system for automatic snapshot selection and zone creation. The experience analytics system receives a request (e.g., selection of a "one-click zoning" button) from a member client device to determine zoning metrics for a webpage based on a selected device type and/or time period. In response, the system accesses plural session replays corresponding to different visitor sessions by one or more customer client devices. The system selects, from among the plural session replays, a session replay based on maximizing scroll amount associated with the session replay. The system then selects, from among plural available snapshots corresponding to the selected session replay, a snapshot based on minimizing display of additional elements (e.g., a menu, a pop-up window, banner or some other element) on the webpage. The system automatically identifies, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage, and determines, based on automatically identifying the zones, zoning metrics for display on the first device.

(13) By virtue of performing automatic snapshot selection and zone creation in this manner, the system provides for facilitated user engagement with respect to zoning. For example, the system reduces manual aspects (e.g., user input for snapshot selection, zone creation and the like) associated with conventional techniques for zone creation. It is therefore possible for the system to perform "one-click zoning." Thus, the system facilitates the creation of zones and the display of corresponding zone metrics, thereby saving time for end users, and reducing computational resources/processing power.

(14) Networked Computing Environment

(15) FIG. **1** is a block diagram showing an example experience analytics system **100** that analyzes and quantifies the user experience of users navigating a client's website, mobile websites, and applications. The experience analytics system **100** can include multiple instances of a member client device **102**, multiple instances of a customer client device **106**, and multiple instances of a third-party server **108**.

(16) The member client device **102** is associated with a client of the experience analytics system **100**, where the client that has a website hosted on the client's third-party server **108**. For example, the client can be a retail store that has an online retail website that is hosted on a third-party server **108**. An agent of the client (e.g., a web administrator, an employee, etc.) can be the user of the member client device **102**.

(17) Each of the member client devices **102** hosts a number of applications, including an experience analytics client **104**. Each experience analytics client **104** is communicatively coupled with an experience analytics server system **124** and third-party servers **108** via a network **110** (e.g., the Internet). An experience analytics client **104** can also communicate with locally-hosted applications using Applications Program Interfaces (APIs).

(18) The member client devices **102** and the customer client devices **106** can also host a number of applications including Internet browsing applications (e.g., Chrome, Safari, etc.). The experience analytics client **104** can also be implemented as a platform that is accessed by the member client device **102** via an Internet browsing application or implemented as an extension on the Internet browsing application.

(19) Users of the customer client device **106** can access client's websites that are hosted on the third-party servers **108** via the network **110** using the Internet browsing applications. For example, the users of the customer client device **106** can navigate to a client's online retail website to purchase goods or services from the website. While the user of the customer client device **106** is navigating the client's website on an Internet browsing application, the Internet browsing application on the customer client device **106** can also execute a client-side script (e.g., JavaScript (.*js)) such as an experience analytics script **122**. In one example, the experience analytics script **122** is hosted on the third-party server **108** with the client's website and processed by the Internet browsing application on the customer client device **106**. The experience analytics script **122** can incorporate a scripting language (e.g., a .*js file or a .json file).

(20) In certain examples, a client's native application (e.g., ANDROID™ or IOS™ Application) is downloaded on the customer client device **106**. In this example, the client's native application including the experience analytics script **122** is programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the experience analytics server system **124**. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the client's native application.

(21) In one example, the experience analytics script **122** records data including the changes in the interface of the website being displayed on the customer client device **106**, the elements on the website being displayed or visible on the interface of the customer client device **106**, the text inputs by the user into the website, a movement of a mouse (or touchpad or touch screen) cursor and mouse (or touchpad or touch screen) clicks on the interface of the website, etc. The experience analytics script **122** transmits the data to experience analytics server system **124** via the network **110**. In another example, the experience analytics script **122** transmits the data to the third-party server **108** and the data can be transmitted from the third-party server **108** to the experience analytics server system **124** via the network **110**.

(22) An experience analytics client **104** is able to communicate and exchange data with the experience analytics server system **124** via the network **110**. The data exchanged between the experience analytics client **104** and the experience analytics server system **124**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., website data, texts reporting

errors, insights, merchandising information, adaptability information, images, graphs providing visualizations of experience analytics, session replay videos, zoning and overlays to be applied on the website, etc.).
(23) The experience analytics server system **124** supports various services and operations that are provided to the experience analytics client **104**. Such operations include transmitting data to and receiving data from the experience analytics client **104**. Data exchanges to and from the experience analytics server system **124** are invoked and controlled through functions available via user interfaces (UIs) of the experience analytics client **104**.
(24) The experience analytics server system **124** provides server-side functionality via the network **110** to a particular experience analytics client **104**. While certain functions of the experience analytics system **100** are described herein as being performed by either an experience analytics client **104** or by the experience analytics server system **124**, the location of certain functionality either within the experience analytics client **104** or the experience analytics server system **124** may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the experience analytics server system **124** but to later migrate this technology and functionality to the experience analytics client **104** where a member client device **102** has sufficient processing capacity.
(25) Turning now specifically to the experience analytics server system **124**, an Application Program Interface (API) server **114** is coupled to, and provides a programmatic interface to, application servers **112**. The application servers **112** are communicatively coupled to a database server **118**, which facilitates access to a database **300** that stores data associated with experience analytics processed by the application servers **112**. Similarly, a web server **120** is coupled to the application servers **112**, and provides web-based interfaces to the application servers **112**. To this end, the web server **120** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.
(26) The Application Program Interface (API) server **114** receives and transmits message data (e.g., commands and message payloads) between the member client device **102** and the application servers **112**. Specifically, the Application Program Interface (API) server **114** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the experience analytics client **104** or the experience analytics script **122** in order to invoke functionality of the application servers **112**. The Application Program Interface (API) server **114** exposes to the experience analytics client **104** various functions supported by the application servers **112**, including generating information on errors, insights, merchandising information, adaptability information, images, graphs providing visualizations of experience analytics, session replay videos, zoning and overlays to be applied on the website, etc.
(27) The application servers **112** host a number of server applications and subsystems, including for example an experience analytics server **116**. The experience analytics server **116** implements a number of data processing technologies and functions, particularly related to the aggregation and other processing of data including the changes in the interface of the website being displayed on the customer client device **106**, the elements on the website being displayed or visible on the interface of the customer client device **106**, the text inputs by the user into the website, a movement of a mouse (or touchpad) cursor and mouse (or touchpad) clicks on the interface of the website, etc. received from multiple instances of the experience analytics script **122** on customer client devices **106**. The experience analytics server **116** implements processing technologies and functions, related to generating user interfaces including information on errors, insights, merchandising information, adaptability information, images, graphs providing visualizations of experience analytics, session replay videos, zoning and overlays to be applied on the website, feedback provided by the user into feedback forms or widgets on the website, etc. Other processor and memory intensive processing of data may also be performed server-side by the experience analytics server **116**, in view of the hardware requirements for such processing.

(28) System Architecture

(29) FIG. **2** is a block diagram illustrating further details regarding the experience analytics system **100** according to some examples. Specifically, the experience analytics system **100** is shown to comprise the experience analytics client **104** and the experience analytics server **116**. The experience analytics system **100** embodies a number of subsystems, which are supported on the client-side by the experience analytics client **104** and on the server-side by the experience analytics server **116**. These subsystems include, for example, a data management system **202**, a data analysis system **204**, a zoning system **206**, a session replay system **208**, a journey system **210**, a merchandising system **212**, an adaptability system **214**, an insights system **216**, an errors system **218**, and an application conversion system **220**.

(30) The data management system **202** is responsible for receiving functions or data from the member client devices **102**, the experience analytics script **122** executed by each of the customer client devices **106**, and the third-party servers **108**. The data management system **202** is also responsible for exporting data to the member client devices **102** or the third-party servers **108** or between the systems in the experience analytics system **100**. The data management system **202** is also configured to manage the third-party integration of the functionalities of experience analytics system **100**.

(31) The data analysis system **204** is responsible for analyzing the data received by the data management system **202**, generating data tags, performing data science and data engineering processes on the data.

(32) The zoning system **206** is responsible for generating a zoning interface to be displayed by the member client device **102** via the experience analytics client **104**. The zoning interface provides a visualization of how the users via the customer client devices **106** interact with each element on the client's website. The zoning interface can also provide an aggregated view of in-page behaviors by the users via the customer client device **106** (e.g., clicks, scrolls, navigation). The zoning interface can also provide a side-by-side view of different versions of the client's website for the client's analysis. For example, the zoning system **206** can identify the zones in a client's website that are associated with a particular element in displayed on the website (e.g., an icon, a text link, etc.). Each zone can be a portion of the website being displayed. The zoning interface can include a view of the client's website. The zoning system **206** can generate an overlay including data pertaining to each of the zones to be overlaid on the view of the client's website. The data in the overlay can include, for example, the number of views or clicks associated with each zone of the client's website within a period of time, which can be established by the user of the member client device **102**. In one example, the data can be generated using information from the data analysis system **204**.

(33) The session replay system **208** is responsible for generating the session replay interface to be displayed by the member client device **102** via the experience analytics client **104**. The session replay interface includes a session replay that is a video reconstructing an individual user's session (e.g., visitor session) on the client's website. The user's session starts when the user arrives into the client's website and ends upon the user's exit from the client's website. A user's session when visiting the client's website on a customer client device **106** can be reconstructed from the data received from the user's experience analytics script **122** on customer client devices **106**. The session replay interface can also include the session replays of a number of different visitor sessions to the client's website within a period of time (e.g., a week, a month, a quarter, etc.). The session replay interface allows the client via the member client device **102** to select and view each of the session replays. In one example, the session replay interface can also include an identification of events (e.g., failed conversions, angry customers, errors in the website, recommendations or insights) that are displayed and allow the user to navigate to the part in the session replay corresponding to the events such that the client can view and analyze the event.

(34) The journey system **210** is responsible for generating the journey interface to be displayed by

the member client device **102** via the experience analytics client **104**. The journey interface includes a visualization of how the visitors progress through the client's website, page-by-page, from entry onto the website to the exit (e.g., in a session). The journey interface can include a visualization that provides a customer journey mapping (e.g., sunburst visualization). This visualization aggregates the data from all of the visitors (e.g., users on different customer client devices **106**) to the website and illustrates the visited pages in the order in which the pages were visited. The client viewing the journey interface on the member client device **102** can identify anomalies such as looping behaviors and unexpected drop-offs. The client viewing the journey interface can also assess the reverse journeys (e.g., pages visitors viewed before arriving at a particular page). The journey interface also allows the client to select a specific segment of the visitors to be displayed in the visualization of the customer journey.

(35) The merchandising system **212** is responsible for generating the merchandising interface to be displayed by the member client device **102** via the experience analytics client **104**. The merchandising interface includes merchandising analysis that provides the client with analytics on: the merchandise to be promoted on the website, optimization of sales performance, the items in the client's product catalog on a granular level, competitor pricing, etc. The merchandising interface can, for example, comprise graphical data visualization pertaining to product opportunities, category, brand performance, etc. For instance, the merchandising interface can include the analytics on conversions (e.g., sales, revenue) associated with a placement or zone in the client website.

(36) The adaptability system **214** is responsible for creating accessible digital experiences for the client's website to be displayed by the customer client devices **106** for visitors that would benefit from an accessibility-enhanced version of the client's website. For instance, the adaptability system **214** can improve the digital experience for users with disabilities, such as visual impairments, cognitive disorders, dyslexia, and age-related needs. The adaptability system **214** can, with proper user permissions, analyze the data from the experience analytics script **122** to determine whether an accessibility-enhanced version of the client's website is needed, and can generate the accessibility-enhanced version of the client's website to be displayed by the customer client device **106**.

(37) The insights system **216** is responsible for analyzing the data from the data management system **202** and the data analysis system **204** surface insights that include opportunities as well as issues that are related to the client's website. The insights can also include alerts that notify the client of deviations from a client's normal business metrics. The insights can be displayed by the member client devices **102** via the experience analytics client **104** on a dashboard of a user interface, as a pop-up element, as a separate panel, etc. In this example, the insights system **216** is responsible for generating an insights interface to be displayed by the member client device **102** via the experience analytics client **104**. In another example, the insights can be incorporated in another interface such as the zoning interface, the session replay, the journey interface, or the merchandising interface to be displayed by the member client device **102**.

(38) The errors system **218** is responsible for analyzing the data from the data management system **202** and the data analysis system **204** to identify errors that are affecting the visitors to the client's website and the impact of the errors on the client's business (e.g., revenue loss). The errors can include the location within the user journey in the website and the page that adversely affects (e.g., causes frustration for) the users (e.g., users on customer client devices **106** visiting the client's website). The errors can also include causes of looping behaviors by the users, in-page issues such as unresponsive calls to action and slow loading pages, etc. The errors can be displayed by the member client devices **102** via the experience analytics client **104** on a dashboard of a user interface, as a pop-up element, as a separate panel, etc. In this example, the errors system **218** is responsible for generating an errors interface to be displayed by the member client device **102** via the experience analytics client **104**. In another example, the insights can be incorporated in another interface such as the zoning interface, the session replay, the journey interface, or the

merchandising interface to be displayed by the member client device **102**.
(39) The application conversion system **220** is responsible for the conversion of the functionalities of the experience analytics server **116** as provided to a client's website to a client's native mobile applications. For instance, the application conversion system **220** generates the mobile application version of the zoning interface, the session replay, the journey interface, the merchandising interface, the insights interface, and the errors interface to be displayed by the member client device **102** via the experience analytics client **104**. The application conversion system **220** generates an accessibility-enhanced version of the client's mobile application to be displayed by the customer client devices **106**.
(40) The feedback system **222** is responsible for receiving and analyzing data from the data management system **202** that includes the feedback data received from the client devices. As the visitor progresses through a client's website on the client device, a feedback webpage of the website, a pop-up window or tab, or an overlay can be displayed to receive the visitor's feedback. For instance, a feedback form can be displayed in a pop-up window or tab of the website, an overlay of the website, one of the plurality of webpages of the website, etc. The visitor can provide feedback on, for example, the functionality of the website, aesthetics of the website, on the goods and services associated with the website, etc. The feedback data can include a text input that is included into a feedback form on the website. The feedback data can also include a survey response, a rating that includes an image, an emoticon, or an icon, a screenshot of one of the plurality of webpages, etc. The feedback system **222** is also responsible for generating feedback interfaces to be displayed by the member client device **102** via the experience analytics client **104**.
(41) As discussed further below with respect to FIG. **6**, the zoning system **206** may include an automatic snapshot selection module **614** and an automatic zone creation module **616**. In response to a user request at the member client device **102** to perform one-click zoning, the automatic snapshot selection module **614** is configured to automatically select a snapshot (e.g., a preferred snapshot) from prior session replays, and the automatic zone creation module **616** is configured to automatically create zones based on the snapshot selected by the automatic snapshot selection module **614**.
(42) Data Architecture
(43) FIG. **3** is a schematic diagram illustrating database **300**, which may be stored in the database **300** of the experience analytics server **116**, according to certain examples. While the content of the database **300** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).
(44) The database **300** includes a data table **302**, a session table **304**, a zoning table **306**, an error table **310**, an insights table **312**, a merchandising table **314**, and a journeys table **308**.
(45) The data table **302** stores data regarding the websites and native applications associated with the clients of the experience analytics system **100**. The data table **302** can store information on the contents of the website or the native application, the changes in the interface of the website being displayed on the customer client device **106**, the elements on the website being displayed or visible on the interface of the customer client device **106**, the text inputs by the user into the website, a movement of a mouse (or touchpad or touch screen) cursor and mouse (or touchpad or touch screen) clicks on the interface of the website, etc. The data table **302** can also store data tags and results of data science and data engineering processes on the data. The data table **302** can also store information such as the font, the images, the videos, the native scripts in the website or applications, etc.
(46) The session table **304** stores session replays for each of the client's websites and native applications.
(47) The zoning table **306** stores data related to the zoning for each of the client's websites and native applications including the zones to be created and the zoning overlay associated with the websites and native applications.

(48) The journeys table **308** stores data related to the journey of each visitor to the client's website or through the native application.
(49) The error table **310** stores data related to the errors generated by the errors system **218** and the insights table **312** stores data related to the insights generated by the insights table **312**.
(50) The merchandising table **314** stores data associated with the merchandising system **212**. For example, the data in the merchandising table **314** can include the product catalog for each of the clients, information on the competitors of each of the clients, the data associated with the products on the websites and applications, the analytics on the product opportunities and the performance of the products based on the zones in the website or application, etc.
(51) The feedback table **316** stores data associated with the feedback system **222**. For example, the data in the feedback table **316** can include the feedback data received from each of the customer client devices **106** and stored in association with the customer client device **106** and the website associated with the customer client device **106**. The feedback data can include, for example, the text input that provides the visitor's (or customer's) feedback on the website, survey response, rating that includes an image, an emoticon, or an icon, a screenshot of one of the plurality of webpages, etc.
(52) FIG. **4** illustrates an unlabeled document object model (DOM) tree **400**, in accordance with some examples. In one or more embodiments, the unlabeled DOM tree **400** provides a visual representation of the hierarchical structure of a webpage's HTML code, with content zones or elements (e.g., as defined by the zoning system **206**) represented as nodes A-F.
(53) In the example unlabeled DOM tree **400**, related nodes A-F are joined by links **402-410**, representing the relationships between any two of the nodes A-F. In the example unlabeled DOM tree **400**, a link **402** is established between nodes A and B, a link **404** is established between nodes A and C, a link **406** is established between nodes B and D, a link **408** is established between nodes B and E, and a link **410** is established between nodes B and F.
(54) In addition, nodes B and C are disposed on a second tier below the first tier occupied by node A, reflecting a structure in which the content element or zone represented by node A includes the content elements or zones represented by nodes B and C. Moreover, nodes D, E and F are disposed on a third tier below the second tier occupied by node B, reflecting a structure in which the content element or zone represented by node B includes the content elements or zones represented by nodes D, E and F.
(55) FIG. **5** is a user interface **502** for presenting a webpage with performance information for zones in a webpage, in accordance with some examples. As described herein, a “zone” is a webpage feature or element included in the HTML and/or the corresponding DOM of the webpage. As noted above with respect to FIG. **4**, a DOM node is a feature of the DOM, where the DOM node corresponds to a given HTML element or zone. Examples of zones in webpages include, but are not limited to, banner advertisements, product images, clickable buttons, and the like.
(56) Performance metrics refer to various quantifiable factors related to goal achievement. As an example, where a given goal targets a fifteen percent newsletter sign-up rate, a corresponding performance metric may be the percentage of site visitors clicking on a “subscribe to newsletter” button. An example metric of interest is an “average hover time,” describing the average amount of time for which users hover the mouse cursor over given elements of the webpage.
(57) With regard to zones (e.g., zone **504**, **506**, **508**, **510**), zoning metrics may be overlaid on the zones for ease of comparison between zones. The zoning metrics include hover rate, click recurrence, attractiveness rate, exposure rate, and exposure time, but other zoning metrics may also be included.
(58) The hover rate is an average time spent hovering over the zone. The click recurrence is the average number of clicks on the zone for page views with at least one click on the zone. The attractiveness rate is the percentage of page views where the zone was visible on the screen with at least one click on the zone. The exposure rate is the percentage of page views where at least half of

the zone was visible on the screen, and the exposure rate indicates how far the users are scrolling. Further, the exposure time is the average time with at least half of the zone is visible on the screen and indicates how long the zone is visible.

(59) In the illustrated example in FIG. **5**, the click recurrence for each zone is shown over imposed over the zone. For example, zone **504** shows that 3.17% of the users that view the zone click on the zone. Zones may also include other zones within, such as zone **506** that includes zone **508** (click recurrence of 0.22%) and zone **510** (click recurrence of 1.74%).

(60) Regarding the exposure rate, in some example embodiments, the test of whether a user views the zone is that the user views at least a threshold portion of the zone. For example, viewing the zone may correspond with viewing the top half of the zone, corresponding to the vertical middle point of the zone is exposed to the user.

(61) FIG. **6** illustrates an architecture **600** for automatic snapshot selection and zone creation, in accordance with some examples. For explanatory purposes, the architecture **600** is primarily described herein with reference to the member client device **102**, the customer client device **106** and the experience analytics server **116** of FIG. **1**. However, the architecture **600** may correspond to one or more other components and/or other suitable devices.

(62) In the example of FIG. **6**, a user (e.g., customer) at the customer client device **106** accesses a website including a webpage **602**. The user interacts with the webpage **602**, with such interactions corresponding to session events performed with respect to the webpage **602**. For example, the session events may include entering text in a text box, clicking a button with a mouse, tapping a button with a touchscreen, scrolling up or down on the webpage **602**, hovering over a webpage element, and the like.

(63) In one or more embodiments, the experience analytics script **122** of the customer client device **106** is configured to track the session events. For example, the experience analytics script **122** may be implemented in part as a tracking tag for the webpage **602**, for tracking the session events within the webpage **602**.

(64) At operation **606**, the customer client device **106** provides the session events to a pipeline, for example, in a serialized format. The experience analytics server **116** is configured to receive the serialized session events, and to store the session events in the database **300** (e.g., within the session table **304**).

(65) In the example of FIG. **6**, operation **606** corresponds to a first phase which relates to storing session events (e.g., user interactions) for the webpage **602**. Moreover, operations **608-612** correspond to a second phase which relates to presenting corresponding zoning metrics **604** (e.g., hover rate, click recurrence, attractiveness rate, exposure rate and/or exposure time for each zone) based on the session events of the webpage **602**. It may be understood that the second phase may occur shortly after the first phase, or after an extended period of time after the first phase.

(66) Regarding the second phase, a member user at the member client device **102** may request (e.g., via user input) to view the zoning metrics **604** for the webpage (e.g., as overlays on respective zones as shown in the example of FIG. **5**). In example embodiments, the member client device **102** displays a user interface which allows the member user to select a webpage of the website (e.g., via a drop-down menu which lists the webpages of the website). Moreover, the member client device **102** displays a user interface with user-selectable elements to specify a time period for determining the zoning metrics **604**. For example, the member client device **102** provides for the user to select a time period corresponding to one of: the last day, the last week, the last month, or a specified date range.

(67) In addition, the member client device **102** displays a user interface which allows the member user to select a device type, including a single device (e.g., desktop, mobile or tablet) or a combination of devices (e.g., selected from desktop, mobile and/or tablet). For example, the user interface is a scrollable list from which the user can select the device type. In example embodiments, the user interface automatically defaults (e.g., preselects) the type of device used in

the most recent zoning session (e.g., a device previously selected by a member client device **102** for zoning metrics).
(68) The member client device **102** further includes a user-selectable button to perform “one-click zoning” based on an identified webpage, the selected device type and/or selected time period (e.g., one or more of which may correspond to default values). As described herein, one-click zoning provides for the zoning system **206** to (a) automatically select a snapshot, from among available snapshots corresponding to a selected session replay stored in the database **300**, and (b) automatically create zones and zoning metrics based on the selected snapshot. As noted above, a session replay corresponds replaying a prior visitor session for a period of time, and is based on accessing session events stored in the database **300**.
(69) Thus, the member client device **102** provides a one-click zoning request (e.g., which identifies a webpage, a selected device type and/or a selected time period) to the experience analytics server **116** for the zoning metrics **604** (operation **608**). In response to the receiving the request, the zoning system **206** of the experience analytics server **116** accesses the session replays stored in the database **300** (operation **610**).
(70) As noted above, the database **300** stores serialized session events. The experience analytics server **116** is configured to unserialize the session events in order to generate the session replays, and to select a suitable session replay from the available session replays. From among the available snapshots in a selected session replay, the automatic snapshot selection module **614** provides for automatically selecting a snapshot (e.g., a preferred snapshot) from which to determine/create zones and to display zoning metrics.
(71) In example embodiments, the automatic snapshot selection module **614** is configured to select a fitting/suitable session replay using multiple requests: attempting to select a session replay with the most (e.g., maximum) scroll that fits the analysis context. For example, the analysis context may relate to one or more of: date range, device, and/or user segment.
(72) If none of the initial session replays fit the analysis context, the automatic snapshot selection module **614** is configured to expand the scroll range (e.g., going down to a predefined percentage, such as 70% scroll) and/or to expand the time period (e.g., expanding a predefined amount beyond the previously-selected last day, week, month, or specified date range), in order to make more session replays available for selection. In this manner, the automatic snapshot selection module **614** may perform up to three different requests.
(73) After selecting a session replay, the automatic snapshot selection module **614** is configured to identify a preferred (e.g., ideal) moment of the user session to capture the snapshot (e.g., by prioritizing “recording events”) within the session replay. In example embodiments, the automatic snapshot selection module **614** attempts to select a moment right before user actions that may change the website layout (e.g., a click). For example, it may be undesirable to select a snapshot in which an additional element (e.g., a menu, a pop-up window, banner or some other element) is displayed. When the automatic snapshot selection module **614** detects such an event, the automatic snapshot selection module **614** is configured to capture a snapshot right before it. In a case where such an event is not detected, the automatic snapshot selection module **614** is configured to check the scroll (e.g., with a goal to have a snapshot with as much/maximum scroll as possible), with a goal to make sure all resources, such as images, are loaded when capturing the snapshot.
(74) In example embodiments, in response to a user selection to change the device type, the member client device **102** is configured to update the snapshot to correspond to the newly-selected device. In this regard, the member client device **102** in conjunction with the automatic snapshot selection module **614** is configured to store/save snapshots (e.g., in the database **300** within a snapshots data structure of the zoning table **306** or the session table **304**) in association with each device type. In a case where a snapshot was not previously stored, the automatic snapshot selection module **614** is configured to create a new snapshot for the matching device type (e.g., and/or user-selected webpage and time period).

(75) Moreover, as part of the one-click zoning, the automatic zone creation module **616** is configured to automatically create zones based on the current snapshot selected by the automatic snapshot selection module **614**. As noted above, a “zone” is a webpage feature or element included in the HTML and/or the corresponding DOM of the webpage **602**. In example embodiments, the automatic zone creation module **616** is configured to traverse the HTML tree (e.g., the DOM tree), and to determine a target for webpage nodes. As described herein, a “target” refers to an identifier computed for a webpage node, such as an HTML node, or corresponding DOM node. Each target may be represented as a target path. Moreover, target values may be variously-computed based on various webpage properties including, without limitation, tag names, element classes, element order in the DOM, specific attributes, and the like, as well as any combination thereof.
(76) Thus, at each level of the DOM tree, the automatic zone creation module **616** checks the target paths of the zones. In doing so, the automatic zone creation module **616** compares the area of the zone to the surface area of the snapshot. If the area of the zones take less than a threshold percentage (e.g., 10%) of the surface area of the snapshot, the automatic zone creation module **616** stops there and selects these as zones with respect to the snapshot. If the area of the zones are greater or equal to the threshold percentage (e.g., 10%), the automatic zone creation module **616** traverses down to the children zones (one level below in the DOM), and re-performs (e.g., rechecks) the prior step (e.g., threshold percentage of surface area). This is repeated for nested children zones until reaching below the threshold percentage. In this manner, it is possible for the automatic zone creation module **616** to create more meaningful zones. For example, the automatic zone creation module **616** prioritizes the surface area that these zones occupy within the snapshot (e.g., with 10% being an arbitrary threshold percentage).
(77) In example embodiments, after automatic creation of the zones, the experience analytics server **116** in conjunction with the zoning system **206** is further configured to determine zoning metrics **604** for the created zones of the webpage **602**. For each target, the zoning system **206** is configured to determine respective zoning metrics **604** for that target. After determining zoning metrics **604**, the zoning system **206** sends an indication of the zoning metrics **604** to the member client device **102** (operation **612**). In response, the member client device **102** provides for display of the created zones and zoning metrics **604**. For example, the zoning metrics **604** are presented as overlays with respect to a representation of the webpage **602** displayed on the member client device **102**.
(78) While the example of FIG. **6** is depicted as a single customer client device **106**, it is possible that the experience analytics server **116** provides for aggregating session replays (e.g., based on serialized session events) from multiple customer client devices **106** (e.g., based on device type). The aggregated data is stored in the database **300**, and is usable by the zoning system **206** to present zoning metrics **604** with respect to the plural customer client devices **106**.
(79) Thus, the experience analytics system **100** as described herein provides for automatic snapshot selection and zone creation. The experience analytics server **116** receives a request (e.g., selection of a “one-click zoning” button) from a member client device **102** to determine zoning metrics for an identified webpage **602** based on a selected device type and/or time period. In response, the experience analytics server **116** accesses plural session replays corresponding to different visitor sessions by one or more customer client devices **106**. The experience analytics server **116** selects, from among the plural session replays, a session replay based on maximizing scroll amount associated with the session replay. The experience analytics server **116** then selects, from among plural available snapshots corresponding to the selected session replay, a snapshot based on minimizing display of additional elements (e.g., a menu, a pop-up window, banner or some other element) on the webpage. The experience analytics server **116** automatically identifies, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage, and determines, based on automatically identifying the zones, zoning metrics for display on the first device.
(80) By virtue of performing automatic snapshot selection and zone creation in this manner, the

experience analytics system **100** provides for increased user and facilitated engagement with respect to zoning. For example, the experience analytics system **100** system reduces manual aspects (e.g., user input for snapshot selection, zone creation and the like) associated with conventional techniques for zone creation. It is therefore possible for the experience analytics system **100** to perform "one-click zoning." Thus, the system facilitates the creation of zones and the display of corresponding zone metrics, thereby saving time for end users, and reducing computational resources/processing power.

(81) FIG. **7** is a flowchart illustrating a process **700** for automatic snapshot selection and zone creation, in accordance with some examples. For explanatory purposes, the process **700** is primarily described herein with reference to the member client device **102**, the customer client device **106** and the experience analytics server **116** of FIG. **1**. However, one or more blocks (or operations) of the process **700** may be performed by one or more other components, and/or by other suitable devices. Further for explanatory purposes, the blocks (or operations) of the process **700** are described herein as occurring in serial, or linearly. However, multiple blocks (or operations) of the process **700** may occur in parallel or concurrently. In addition, the blocks (or operations) of the process **700** need not be performed in the order shown and/or one or more blocks (or operations) of the process **700** need not be performed and/or can be replaced by other operations. The process **700** may be terminated when its operations are completed. In addition, the process **700** may correspond to a method, a procedure, an algorithm, etc.

(82) The experience analytics server **116** receives, from the member client device **102**, an indication of user input selecting to determine zoning metrics for a webpage based on a selected device type (block **702**). In example embodiments, determining the zoning metrics is performed in association with an analysis context for the webpage.

(83) The experience analytics server **116** accesses, in response to receiving the user input, plural session replays corresponding to different visitor sessions by at least one customer client device **106** with respect to the webpage (**704**). The experience analytics server **116** selects, from among the plural session replays, a session replay based at least in part on maximizing scroll amount associated with the session replay (block **706**). In example embodiments, selecting the session replay is further based on the selected device type.

(84) In example embodiments, the experience analytics server **116** determines that an initial subset of the plural session replays does not include a session replay that matches the analysis context for the webpage based on maximizing the scroll amount, expands, based on the determining, a scroll range for the plural session replays beyond the initial subset, and selects the session replay based on the expanded scroll range. Alternatively or in addition, the experience analytics server **116** determines that an initial subset of the plural session replays does not include a session replay that matches the analysis context for the webpage based on maximizing the scroll amount, expands, based on the determining, a time period range for the plural session replays beyond the initial subset, and selects the session replay based on the expanded time period range.

(85) The experience analytics server **116** selects, from among plural available snapshots corresponding to the selected session replay, a snapshot based at least in part on minimizing display of additional elements on the webpage (block **708**). For example, the additional elements on the webpage comprise at least one of a menu or a pop-up element.

(86) In example embodiments, selecting the snapshot is further based on maximizing scroll amount associated with the snapshot. Alternatively or in addition, selecting the snapshot is further based on determining that all resources are loaded with respect to the snapshot.

(87) The experience analytics server **116** automatically identifies, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage (block **710**). The experience analytics server **116** determines, based on automatically identifying the zones, zoning metrics for display on the member client device **102** (block **712**).

(88) Machine Architecture

(89) FIG. **8** is a diagrammatic representation of the machine **800** within which instructions **810** (e.g., software, a program, an application, an applet, an application, or other executable code) for causing the machine **800** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **810** may cause the machine **800** to execute any one or more of the methods described herein. The instructions **810** transform the general, non-programmed machine **800** into a particular machine **800** programmed to carry out the described and illustrated functions in the manner described. The machine **800** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **800** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **800** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **810**, sequentially or otherwise, that specify actions to be taken by the machine **800**. Further, while only a single machine **800** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **810** to perform any one or more of the methodologies discussed herein. The machine **800**, for example, may comprise the member client device **102** or any one of a number of server devices forming part of the experience analytics server **116**. In some examples, the machine **800** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.
(90) The machine **800** may include processors **804**, memory **806**, and input/output I/O components **802**, which may be configured to communicate with each other via a bus **840**. In an example, the processors **804** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **808** and a processor **812** that execute the instructions **810**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **8** shows multiple processors **804**, the machine **800** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.
(91) The memory **806** includes a main memory **814**, a static memory **816**, and a storage unit **818**, both accessible to the processors **804** via the bus **840**. The main memory **814**, the static memory **816**, and storage unit **818** store the instructions **810** embodying any one or more of the methodologies or functions described herein. The instructions **810** may also reside, completely or partially, within the main memory **814**, within the static memory **816**, within machine-readable medium **820** within the storage unit **818**, within at least one of the processors **804** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **800**.
(92) The I/O components **802** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **802** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include

such a touch input device. It will be appreciated that the I/O components **802** may include many other components that are not shown in FIG. **8**. In various examples, the I/O components **802** may include user output components **826** and user input components **828**. The user output components **826** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **828** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(93) In further examples, the I/O components **802** may include biometric components **830**, motion components **832**, environmental components **834**, or position components **836**, among a wide array of other components. For example, the biometric components **830** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **832** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

(94) The environmental components **834** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

(95) With respect to cameras, the member client device **102** may have a camera system comprising, for example, front cameras on a front surface of the member client device **102** and rear cameras on a rear surface of the member client device **102**. The front cameras may, for example, be used to capture still images and video of a user of the member client device **102** (e.g., “selfies”). The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode. In addition to front and rear cameras, the member client device **102** may also include a 360° camera for capturing 360° photographs and videos.

(96) Further, the camera system of a member client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the member client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera and a depth sensor, for example.

(97) The position components **836** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

(98) Communication may be implemented using a wide variety of technologies. The I/O components **802** further include communication components **838** operable to couple the machine **800** to a network **822** or devices **824** via respective coupling or connections. For example, the communication components **838** may include a network interface component or another suitable

device to interface with the network **822**. In further examples, the communication components **838** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **824** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

(99) Moreover, the communication components **838** may detect identifiers or include components operable to detect identifiers. For example, the communication components **838** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **838**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

(100) The various memories (e.g., main memory **814**, static memory **816**, and memory of the processors **804**) and storage unit **818** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **810**), when executed by processors **804**, cause various operations to implement the disclosed examples.

(101) The instructions **810** may be transmitted or received over the network **822**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **838**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **810** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **824**.

(102) Software Architecture

(103) FIG. **9** is a block diagram **900** illustrating a software architecture **904**, which can be installed on any one or more of the devices described herein. The software architecture **904** is supported by hardware such as a machine **902** that includes processors **920**, memory **926**, and I/O components **938**. In this example, the software architecture **904** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **904** includes layers such as an operating system **912**, libraries **910**, frameworks **908**, and applications **906**. Operationally, the applications **906** invoke API calls **950** through the software stack and receive messages **952** in response to the API calls **950**.

(104) The operating system **912** manages hardware resources and provides common services. The operating system **912** includes, for example, a kernel **914**, services **916**, and drivers **922**. The kernel **914** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **914** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionalities. The services **916** can provide other common services for the other software layers. The drivers **922** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **922** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

(105) The libraries **910** provide a common low-level infrastructure used by the applications **906**. The libraries **910** can include system libraries **918** (e.g., C standard library) that provide functions

such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **910** can include API libraries **924** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **910** can also include a wide variety of other libraries **928** to provide many other APIs to the applications **906**.

(106) The frameworks **908** provide a common high-level infrastructure that is used by the applications **906**. For example, the frameworks **908** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **908** can provide a broad spectrum of other APIs that can be used by the applications **906**, some of which may be specific to a particular operating system or platform.

(107) In an example, the applications **906** may include a home application **936**, a contacts application **930**, a browser application **932**, a book reader application **934**, a location application **942**, a media application **944**, a messaging application **946**, a game application **948**, and a broad assortment of other applications such as a third-party application **940**. The applications **906** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **906**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **940** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **940** can invoke the API calls **950** provided by the operating system **912** to facilitate functionality described herein.

Glossary

(108) "Carrier signal" refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

(109) "Client device" refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

(110) "Communication network" refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a

Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

(111) "Component" refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A "hardware component" is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase "hardware component" (or "hardware-implemented component") should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over

appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, "processor-implemented component" refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors **1004** or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a "cloud computing" environment or as a "software as a service" (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

(112) "Computer-readable storage medium" refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms "machine-readable medium," "computer-readable medium" and "device-readable medium" mean the same thing and may be used interchangeably in this disclosure.

(113) "Ephemeral message" refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

(114) "Machine storage medium" refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks The terms "machine-storage medium," "device-storage medium," "computer-storage medium" mean the same thing and may be used interchangeably in this disclosure. The terms "machine-storage media," "computer-storage media," and "device-storage media" specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are

covered under the term “signal medium.”
(115) “Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.
(116) “Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a matter as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

Claims

1. A method, comprising: receiving, from a first device, an indication of user input selecting to determine zoning metrics for a webpage based on a selected device type; accessing, in response to receiving the user input, plural session replays corresponding to different visitor sessions by at least one second device with respect to the webpage; selecting, from among the plural session replays, a session replay based at least in part on maximizing scroll amount associated with the session replay; selecting, from among plural available snapshots corresponding to the selected session replay, a snapshot based at least in part on minimizing display of additional elements on the webpage; automatically identifying, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage; determining, based on automatically identifying the zones, zoning metrics for display on the first device, wherein determining the zoning metrics is performed in association with an analysis context for the webpage; determining that an initial subset of the plural session replays does not include a session replay that matches the analysis context for the webpage based on maximizing the scroll amount; expanding, based on the determining, a scroll range for the plural session replays beyond the initial subset; and selecting the session replay based on the expanded scroll range.
2. The method of claim 1, wherein selecting the session replay is further based on the selected device type.
3. The method of claim 1, wherein selecting the snapshot is further based on maximizing scroll amount associated with the snapshot.
4. The method of claim 1, wherein the additional elements on the webpage comprise at least one of a menu or a pop-up element.
5. A method, comprising: receiving, from a first device, an indication of user input selecting to determine zoning metrics for a webpage based on a selected device type; accessing, in response to receiving the user input, plural session replays corresponding to different visitor sessions by at least one second device with respect to the webpage; selecting, from among the plural session replays, a session replay based at least in part on maximizing scroll amount associated with the session replay; selecting, from among plural available snapshots corresponding to the selected session replay, a snapshot based at least in part on minimizing display of additional elements on the webpage; automatically identifying, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage; determining, based on automatically identifying the zones, zoning metrics for display on the first device, wherein determining the zoning metrics is performed in association with an analysis context for the webpage; determining that an initial subset of the plural session replays does not include a session replay that matches the analysis context for the webpage based on maximizing the scroll amount; expanding, based on the determining, a time period range for the plural session replays beyond the initial subset; and selecting the session replay based on the expanded time period range.
6. A system comprising: at least one processor; and a memory storing instructions that, when

executed by the at least one processor, configure the at least one processor to perform operations comprising: receiving, from a first device, an indication of user input selecting to determine zoning metrics for a webpage based on a selected device type; accessing, in response to receiving the user input, plural session replays corresponding to different visitor sessions by at least one second device with respect to the webpage; selecting, from among the plural session replays, a session replay based at least in part on maximizing scroll amount associated with the session replay; selecting, from among plural available snapshots corresponding to the selected session replay, a snapshot based at least in part on minimizing display of additional elements on the webpage; automatically identifying, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage; determining, based on automatically identifying the zones, zoning metrics for display on the first device, wherein determining the zoning metrics is performed in association with an analysis context for the webpage; determining that an initial subset of the plural session replays does not include a session replay that matches the analysis context for the webpage based on maximizing the scroll amount; expanding, based on the determining, a scroll range for the plural session replays beyond the initial subset; and selecting the session replay based on the expanded scroll range.
7. The system of claim 6, wherein selecting the session replay is further based on the selected device type.
8. The system of claim 6, wherein selecting the snapshot is further based on maximizing scroll amount associated with the snapshot.
9. The system of claim 6, wherein the additional elements on the webpage comprise at least one of a menu or a pop-up element.
10. A system comprising: at least one processor; and a memory storing instructions that, when executed by the at least one processor, configure the at least one processor to perform operations comprising: receiving, from a first device, an indication of user input selecting to determine zoning metrics for a webpage based on a selected device type; accessing, in response to receiving the user input, plural session replays corresponding to different visitor sessions by at least one second device with respect to the webpage; selecting, from among the plural session replays, a session replay based at least in part on maximizing scroll amount associated with the session replay; selecting, from among plural available snapshots corresponding to the selected session replay, a snapshot based at least in part on minimizing display of additional elements on the webpage; automatically identifying, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage; determining, based on automatically identifying the zones, zoning metrics for display on the first device, wherein determining the zoning metrics is performed in association with an analysis context for the webpage; determining that an initial subset of the plural session replays does not include a session replay that matches the analysis context for the webpage based on maximizing the scroll amount; expanding, based on the determining, a time period range for the plural session replays beyond the initial subset; and selecting the session replay based on the expanded time period range.
11. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to perform operations comprising: receiving, from a first device, an indication of user input selecting to determine zoning metrics for a webpage based on a selected device type; accessing, in response to receiving the user input, plural session replays corresponding to different visitor sessions by at least one second device with respect to the webpage; selecting, from among the plural session replays, a session replay based at least in part on maximizing scroll amount associated with the session replay; selecting, from among plural available snapshots corresponding to the selected session replay, a snapshot based at least in part on minimizing display of additional elements on the webpage; automatically identifying, based on traversing a tree diagram corresponding to the selected snapshot, zones for webpage; determining, based on automatically identifying the zones, zoning

metrics for display on the first device, wherein determining the zoning metrics is performed in association with an analysis context for the webpage; determining that an initial subset of the plural session replays does not include a session replay that matches the analysis context for the webpage based on maximizing the scroll amount; expanding, based on the determining, a scroll range for the plural session replays beyond the initial subset; and selecting the session replay based on the expanded scroll range.

12. The non-transitory computer-readable storage medium of claim 11, wherein selecting the session replay is further based on the selected device type.

13. The non-transitory computer-readable storage medium of claim 11, wherein selecting the snapshot is further based on maximizing scroll amount associated with the snapshot.